Staged Context Extension + Cooldown

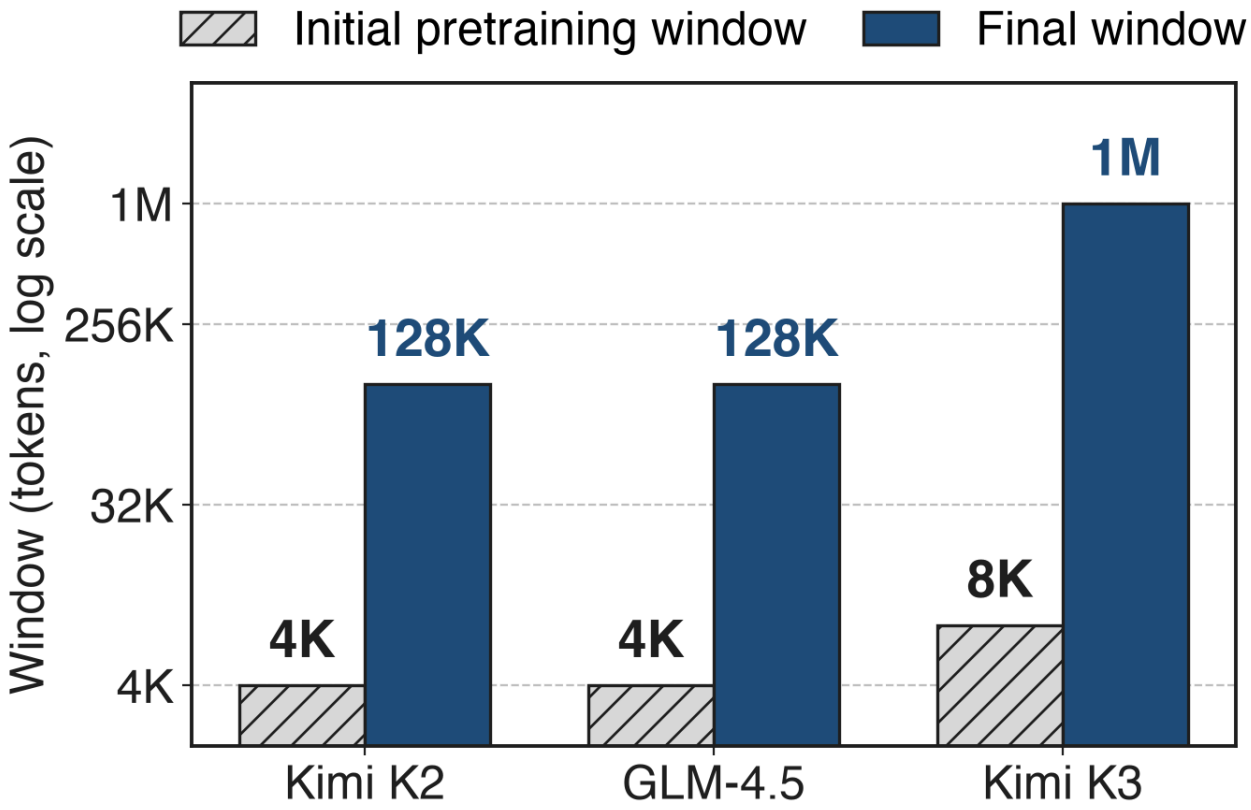
Train short, stretch the window in the low-LR tail

Moonshot AI (Kimi K2, Kimi K3) · Zhipu AI (GLM-4.5)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Frontier MoEs pretrain short, then serve long

- Kimi K2: **1.04T** MoE, **15.5T** tokens
- GLM-4.5: **355B** MoE, **23T** tokens
- Kimi K3: **2.8T** MoE, **1M** window



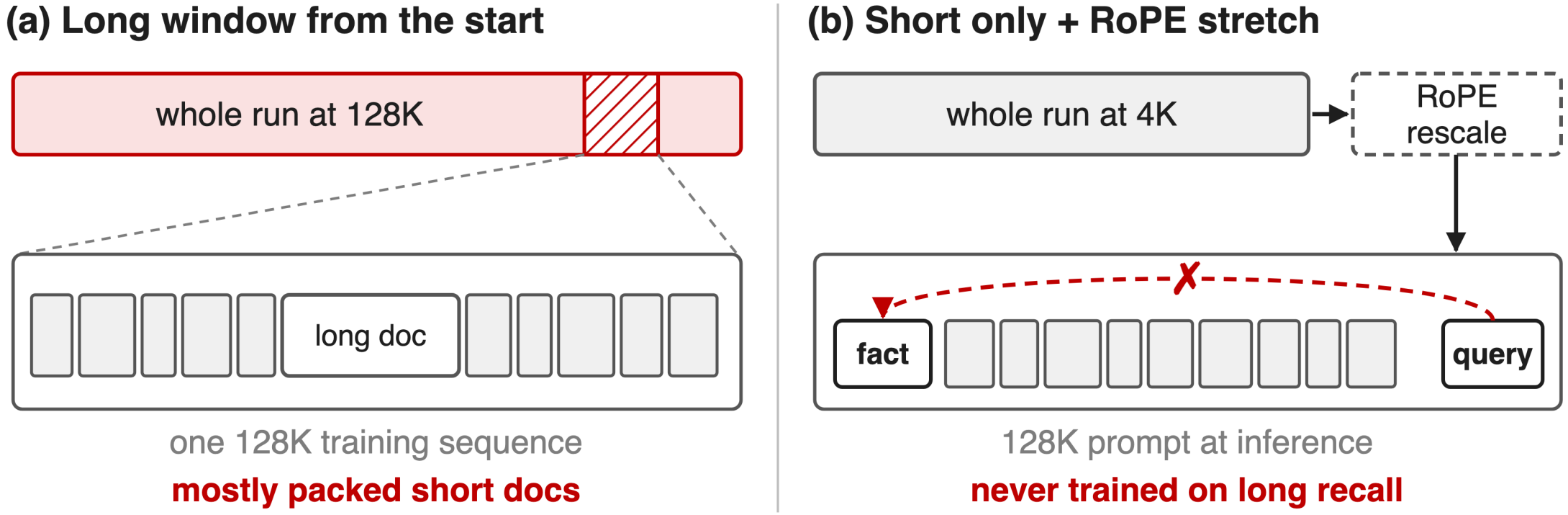
Long windows multiply attention work per token

32×

attention work per token in a 128K window vs. a 4K window (analytic);
128× from 8K to 1M

Analytic: causal attention reads every earlier token, so work per token grows with the window. Windows: K2 §2.5, GLM-4.5 §2.4, K3 §3.4

Always-long and never-long training both fail



(a) pays the long window on every token; (b) never practises using it

Insight: long-range skill is a late, small adaptation

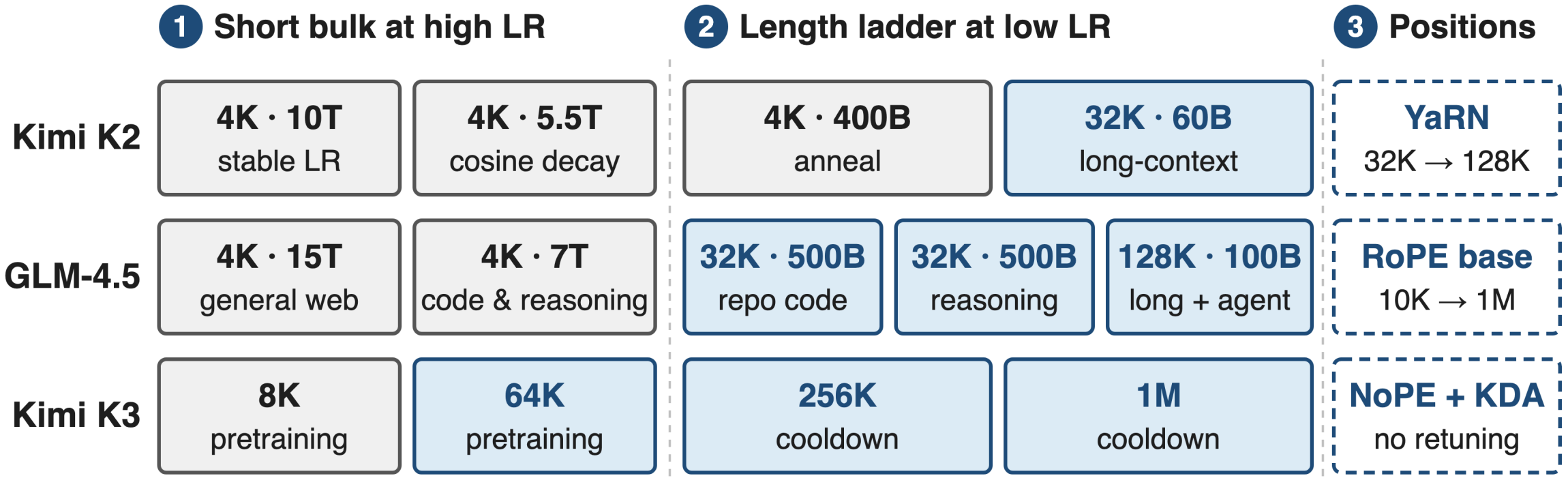
- Knowledge and language: learnable in **4–8K** windows
- Using distant context: needs long **data**, not a long **run**

Insight: long-range skill is a late, small adaptation

- Knowledge and language: learnable in **4–8K** windows
- Using distant context: needs long **data**, not a long **run**

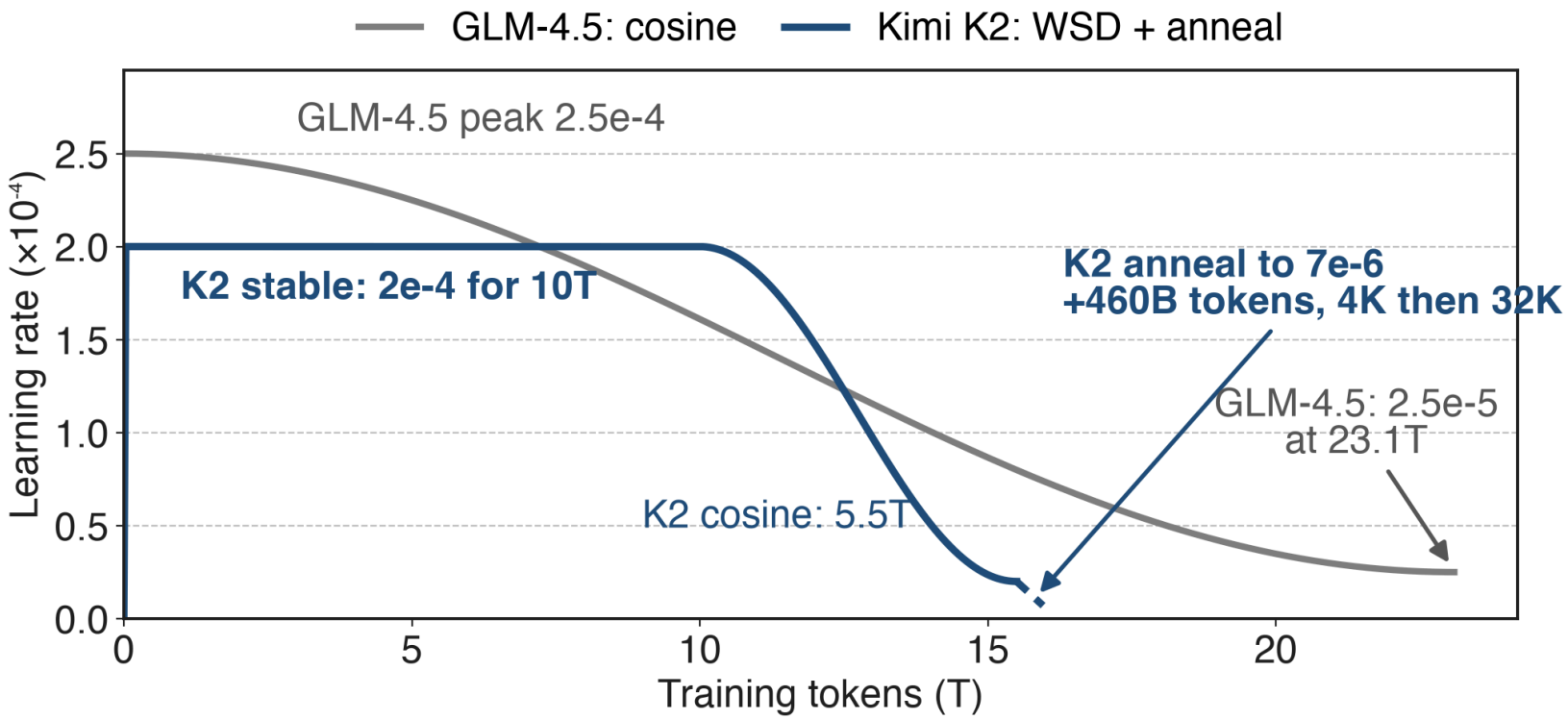
Spend long windows only in the low-LR tail, on data that needs them

Three runs share one shape: short bulk, long tail



Not to scale. Blue = windows above the starting length. K3 publishes stage windows but no token budgets.

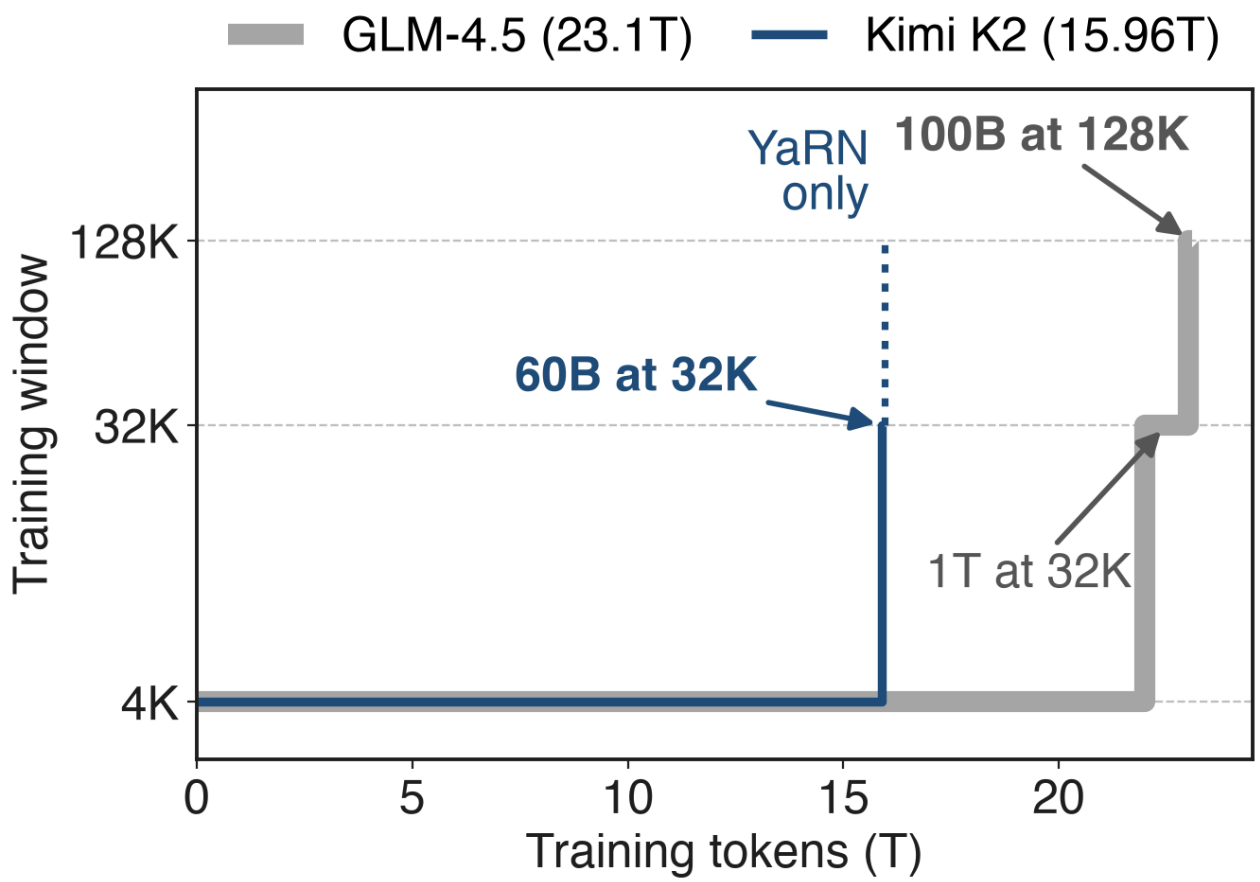
❶ The LR schedule decays into a low-LR tail



GLM-4.5 and K3 chose cosine: WSD underfit, or lost once each was tuned

❷ The length ladder fills the last few percent

- K2: **60B** tokens at 32K = **0.4%**
- GLM-4.5: **1.1T** at ≥ 32K (**4.8%**), **100B** at 128K
- Long docs upsampled; K3 adds synthetic long-range tasks

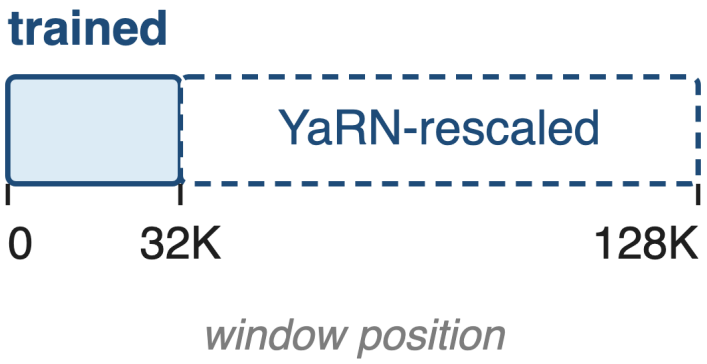


To scale; K3 not plotted (stage budgets unpublished). Shares derived.

❸ Each rung re-fits positions, or needs none

(a) Kimi K2

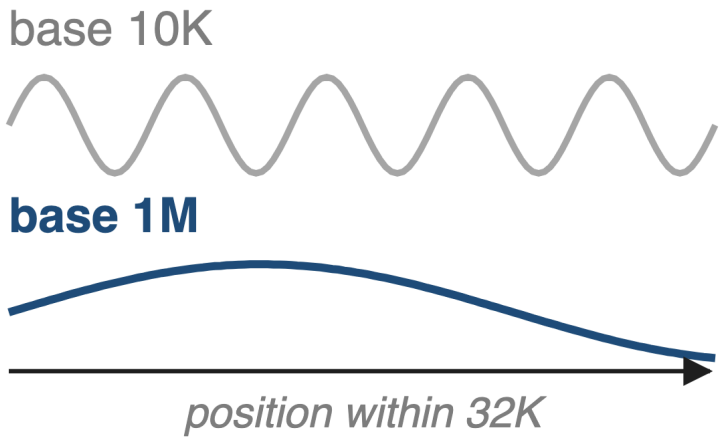
RoPE + YaRN



128K from a 32K stage

(b) GLM-4.5

RoPE base 10K → 1M



slower rotation per position

(c) Kimi K3

NoPE + KDA



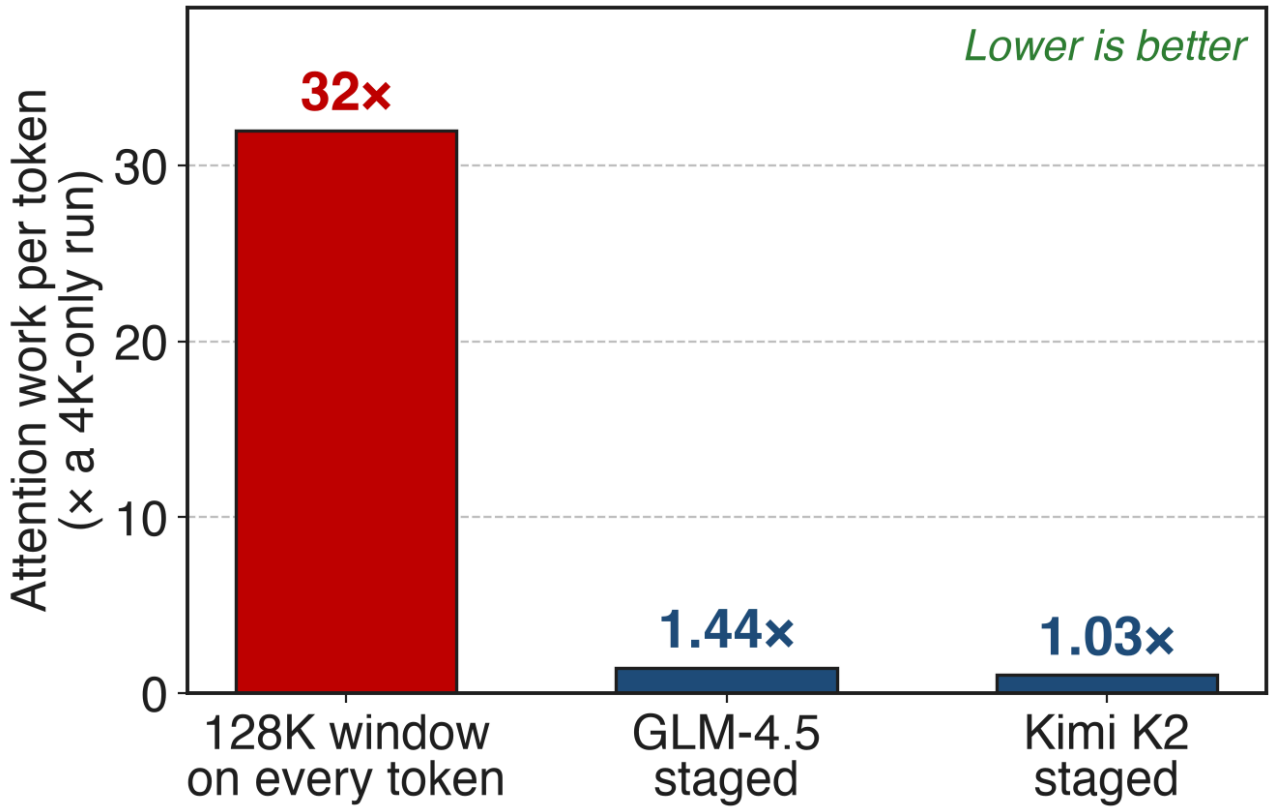
same layers at every window

8K to 1M with no change

K3's KDA layers carry order, so 8K → 1M needs no RoPE change (deck 05)

Staging keeps attention work within 1.44× of 4K

- Pain was **32×** per token at 128K
- Run average: **1.03×** (K2), **1.44×** (GLM-4.5)
- GLM-4.5's 128K rung: **0.4%** of tokens, ~**10%** of attention work



Analytic, token-weighted from published stage budgets

The trained windows hold up end to end

- **Kimi K2**, 128K window
- MRCR **55.0%** vs 50.8% (DeepSeek-V3-0324)

- **Kimi K3**, 1M window
- BrowseComp **90.4%** with no context management
- AA-LCR **74.7**

Whole-recipe outcomes: no report ablates staging alone

The cost: scarce data, extra infra, few ablations

- **Long data is scarce**: upsample, synthesize, clean
- Tail still needs context parallelism (deck 69)
- K2's last 4×: YaRN, not trained
- No staged-vs-unstaged ablation; K3 budgets unpublished
- **Not for RL output length**: GLM-4.5 single 64K beat 16K→64K, **83.4%** vs 80.6%

Summary

- Long windows: **32×** attention work per token (128K vs 4K)
- **Insight**: learn knowledge short, teach length in the low-LR tail
- Top rung: **0.4%** of tokens in K2 and GLM-4.5
- Run average **1.03–1.44×** a 4K run; K3 reaches **1M**

https://arxiv.org/abs/2607.24653